Graph Learning for Interpretable Urban Accessibility and Opportunity

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Cities are complex, interconnected systems where access to opportunities is shaped by spatial structure, infrastructure, and social dynamics. Graph learning offers a powerful framework to model these relationships, enabling interpretable insights into urban accessibility and the distribution of opportunity. The presence of a multimodal data that integrate GTFS transit schedules, census demographics (ACS), employment locations (LEDH), and land-use maps (OSM) allow us to to compute accessibility indices (e.g. number of jobs reachable within 30–60 min by transit or car).

The goal is to train, among others, tree-based graph learning models (Random Forest, XGBoost) to predict these accessibility scores from spatial features (e.g. population density, land-use mix, transit stop density, travel speeds) and then explain the results with SHAP values and partial-dependence plots. Clustering the SHAP “signatures” will reveal zones with similar access patterns (“access deserts”). The explainable outputs support equity-focused planning by showing which factors (e.g. lack of transit stops or mixed-use) most depress access in underserved areas.

Within this framework, the project will address two central research questions:
1. How accurately can graph learning models predict multimodal accessibility indices from spatial and socio-economic features across urban environments?
2. Which spatial factors most strongly influence urban accessibility, and how can interpretable methods (e.g., SHAP values and clustering of feature attributions) reveal distinct patterns of opportunity?

Project offered to study/studies: Bachelor Informatica (computer science), Bachelor Kunstmatige Intelligentie (artificial intelligence)

Max number of students: 2